

Tensor Analysis of fMRI Data

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Outline of Talk

1) Bilinear Models:

- Introduction
- Principal Components
- Independent Components

2) Multimode Models:

- Introduction
- Tucker3 Model
- Parallel Factors Model

3) Application to fMRI Data:

- Introduction to fMRI
- Tensor Probabilistic ICA
- Some Simulation Results

Bilinear Models

Data Matrix and Mean-Centering

Let $\mathbf{X} = \{x_{ij}\}_{I \times J}$ denote a data matrix and assume WLOG $I > J$.

- x_{ij} is signal from i -th voxel at j -th time point

Columnwise mean-centered data matrix has the form

$$\mathbf{X}_c = (\mathbf{I}_I - I^{-1} \mathbf{1}_I \mathbf{1}_I') \mathbf{X}$$

where

- $\mathbf{I}_I$ is the $I \times I$ identity matrix
- $\mathbf{1}_I$ is an $I \times 1$ vector of ones

In remaining slides, I drop the “c” subscript and assume $\mathbf{X}$ is mean-centered.

Singular Value Decomposition

The **Singular Value Decomposition** (SVD) of $\mathbf{X}$ (Eckart & Young, 1936)

$$\mathbf{X} = \mathbf{U}\mathbf{S}\mathbf{V}'$$

where

- $\mathbf{U} = [\mathbf{u}_1, \dots, \mathbf{u}_{\tilde{R}}]$ are **left singular vectors** such that $\mathbf{U}'\mathbf{U} = \mathbf{I}_{\tilde{R}}$
- $\mathbf{S} = \text{diag}(s_1, \dots, s_{\tilde{R}})$ contains **singular values** with $s_1 \geq \dots \geq s_{\tilde{R}} > 0$
- $\mathbf{V} = [\mathbf{v}_1, \dots, \mathbf{v}_{\tilde{R}}]$ are **right singular vectors** such that $\mathbf{V}'\mathbf{V} = \mathbf{I}_{\tilde{R}}$
- $\tilde{R}$ is the rank of $\mathbf{X}$ (note: $\tilde{R} = \min(I, J)$ if $\mathbf{X}$ is full rank)

$\hat{\mathbf{X}} = \mathbf{U}_R \mathbf{S}_R \mathbf{V}_R'$ is best rank- R approximation of $\mathbf{X}$, i.e., minimizes $\sum (x_{ij} - \hat{x}_{ij})^2$

- $\mathbf{U}_R$ and $\mathbf{V}_R$ contain first R columns of $\mathbf{U}$ and $\mathbf{V}$ (respectively)
- $\mathbf{S}_R = \text{diag}(s_1, \dots, s_R)$ contains the R largest singular values

Bilinear Model Form

A **bilinear model** has the form:

$$x_{ij} = \sum_{r=1}^R a_{ir}b_{jr} + e_{ij} \quad \text{or} \quad \mathbf{X} = \mathbf{AB}' + \mathbf{E}$$

where

- $R \leq J$ is the number of components
- $\mathbf{A} = \{a_{ir}\}_{I \times R}$ are the **component scores**
- $\mathbf{B} = \{b_{jr}\}_{J \times R}$ are the **component weights** (loadings)
- $\mathbf{E} = \{e_{ij}\}_{I \times J}$ are the error terms

Rotational Indeterminacy in Bilinear Models

Suppose that $\mathbf{R}$ is an $R \times R$ orthogonal rotation matrix.

- $\mathbf{R}'\mathbf{R} = \mathbf{R}\mathbf{R}' = \mathbf{I}_R$ where $\mathbf{I}_R$ is $R \times R$ identity matrix

Rotational Indeterminacy Problem:

- $\mathbf{A}\mathbf{B}' = \tilde{\mathbf{A}}\tilde{\mathbf{B}}'$ where $\tilde{\mathbf{A}} = \mathbf{A}\mathbf{R}$ and $\tilde{\mathbf{B}} = \mathbf{B}\mathbf{R}$

Implications of Rotational Indeterminacy:

- $(\mathbf{A}, \mathbf{B})$ and $(\tilde{\mathbf{A}}, \tilde{\mathbf{B}})$ produce same $\hat{\mathbf{X}}$ but can be very different
- Need to make additional assumptions to estimate $\mathbf{A}$ and $\mathbf{B}$

Principal Component Analysis

Principal Component Analysis (PCA) finds orthogonal components that minimize $\sum (x_{ij} - \hat{x}_{ij})^2$, see Pearson (1901) and Hotelling (1933).

Principal Component Analysis Assumptions:

- Columns of $\mathbf{A}$ are orthogonal
- Columns of $\mathbf{B}$ are orthonormal
- $d_1 \geq \dots \geq d_R$ where $(d_1, \dots, d_R)' = \text{diag}(\mathbf{A}'\mathbf{A})$

Assuming that columns of $\mathbf{X}$ are mean-centered, PCA solution is

- $\hat{\mathbf{A}} = \mathbf{U}_R \mathbf{S}_R$ and $\hat{\mathbf{B}} = \mathbf{V}_R$ where $\hat{\mathbf{X}} = \mathbf{U}_R \mathbf{S}_R \mathbf{V}_R'$ is rank- R SVD
- Note: $\hat{\mathbf{B}}$ is first R eigenvectors of covariance matrix $\mathbf{C} = \frac{1}{I} \mathbf{X}'\mathbf{X}$

Probabilistic Principal Component Analysis

Probabilistic PCA (Tipping & Bishop, 1999) assumes that

$$\mathbf{x} = \mathbf{B}\mathbf{a} + \mathbf{e}$$

where

- $\mathbf{x} = (x_1, \dots, x_J)$ is observed J -dimensional variable
- $\mathbf{a} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_R)$ is Gaussian latent **source signal** vector
- $\mathbf{B} = \{b_{jr}\}_{J \times R}$ is latent **mixing matrix**
- $\mathbf{e} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_J)$ is Gaussian latent **white noise** vector

Assuming that columns of $\mathbf{X}$ are mean-centered, MLE solution is

- $\hat{\mathbf{A}} = \mathbf{U}_R \mathbf{S}_R (\mathbf{I}^{-1} \mathbf{S}_R^2 - \sigma^2 \mathbf{I}_R)^{-1/2} \mathbf{R}$ where $\mathbf{R}$ is rotation matrix
- $\hat{\mathbf{B}} = \mathbf{V}_R (\mathbf{I}^{-1} \mathbf{S}_R^2 - \sigma^2 \mathbf{I}_R)^{1/2} \mathbf{R}$ where $\mathbf{R}$ is rotation matrix
- $\hat{\sigma}^2 = \frac{1}{I(J-R)} \sum_{j=R+1}^J s_j^2$ where s_j is j -th singular value

Independent Component Analysis

Independent Component Analysis (ICA) is an extension of PCA that finds independent components, see Comon (1994) and De Lathauwer et al. (2000).

Independent Component Analysis Assumptions:

- Columns of $\mathbf{A}$ are independent non-Gaussian **source signals**
- Columns of $\mathbf{B}$ are latent **mixing vectors**

Assuming that columns of $\mathbf{X}$ are mean-centered, ICA solution is

- $\mathbf{X}_w = \mathbf{XV}_R\mathbf{S}_R^{-1}I^{1/2} = I^{1/2}\mathbf{U}_R$ is the **whitened** version of the data
- $\hat{\mathbf{A}} = \mathbf{X}_w\mathbf{R}$ and $\hat{\mathbf{B}} = I^{-1/2}\mathbf{V}_R\mathbf{S}_R\mathbf{R}$ where $\mathbf{R}$ is rotation matrix

Probabilistic Independent Component Analysis

Probabilistic ICA (see Penny et al., 2001; Beckmann & Smith, 2004) assumes

$$\mathbf{x} = \mathbf{B}\mathbf{a} + \mathbf{e}$$

where

- $\mathbf{x} = (x_1, \dots, x_J)$ is observed J -dimensional variable
- $\mathbf{a} \sim (\mathbf{0}, \mathbf{I}_R)$ is non-Gaussian latent **source signal** vector
- $\mathbf{B} = \{b_{jr}\}_{J \times R}$ is latent **mixing matrix**
- $\mathbf{e} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_J)$ is Gaussian latent **white noise** vector

Assuming that columns of $\mathbf{X}$ are mean-centered, PICA solution is

- $\mathbf{X}_q = \mathbf{U}_R \mathbf{S}_R (I^{-1} \mathbf{S}_R^2 - \sigma^2 \mathbf{I}_R)^{-1/2}$ is the **quasiwhitened** version of the data
- $\hat{\mathbf{A}} = \mathbf{X}_q \mathbf{R}$ and $\hat{\mathbf{B}} = \mathbf{V}_R (I^{-1} \mathbf{S}_R^2 - \sigma^2 \mathbf{I}_R)^{1/2} \mathbf{R}$
- $\hat{\sigma}^2 = \frac{1}{I(J-R)} \sum_{j=R+1}^J s_j^2$ where s_j is j -th singular value

Approximating Statistical Independence

Many algorithms exist for estimating the optimal rotation matrix $\mathbf{R}$.

Information Maximization (InfoMax) (Bell & Sejnowski, 1995)

- Maximizes output entropy via nonlinear “squashing” function

Fast ICA (Hyvärinen, 1999) :

- Maximizes negentropy via nonquadratic contrast function

Joint Approximate Diagonalization of Eigenmatrices (JADE)

(Cardoso & Souloumiac, 1993, 1996) :

- Diagonalizes 4-th order cumulant array via Jacobi rotations

Multimode Models

Historical Development of Multimode Models

Cattell's (1944) **Principle of Parallel Proportional Profiles**:

- Attempted to resolve rotational indeterminacy in bilinear model
- “Real” components should have an invariant structure across different conditions, and should vary proportionally depending on the relative importance of each component in each condition

Tucker (1966) proposed first 3-way factor model (Tucker3 Model).

- Very useful model, but doesn't resolve rotational indeterminacy

Idea of trilinear variation was formalized independently by

- Harshman's (1970) Parallel Factors (PARAFAC) Model
- Carroll and Chang's (1970) Canonical Decomposition Model

Some Multimode References

Useful Books:

- *Research Methods for Multimode Data Analysis* (1984)
- *Multi-way Analysis with Applications in the Chemical Sciences* (2004)
- *Applied Multiway Data Analysis* (2008)

Useful Articles:

- *Tensor decompositions and applications.* (Kolda & Bader, 2009). SIAM.
- *PARAFAC: Parallel Factor Analysis.* (Harshman & Lundy, 1994). CSDA.
- *Three-Way Component Analysis Using the R Package ThreeWay.* (Giordani et al., 2014). JSS.

Tucker's Three-Mode Factor Analysis Model

Let $\mathbf{X} = \{x_{ijk}\}_{I \times J \times K}$ denote a three-way data array.

- x_{ijk} is signal from i -th voxel at j -th time point of k -th subject

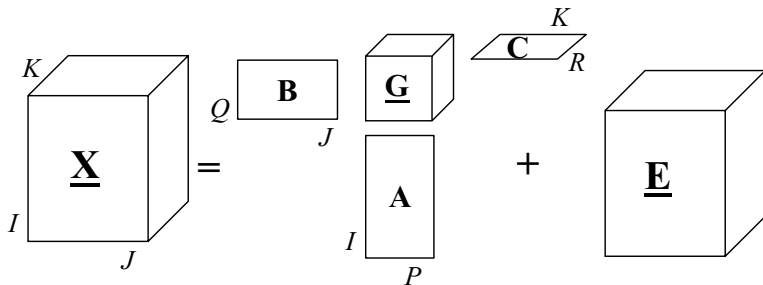
Tucker's (1966) **three-mode factor analysis model** has the form

$$x_{ijk} = \sum_{p=1}^P \sum_{q=1}^Q \sum_{r=1}^R a_{ip} b_{jq} c_{kr} g_{pqr} + e_{ijk}$$

where

- $\mathbf{A} = \{a_{ip}\}_{I \times P}$ are the Mode A weights
- $\mathbf{B} = \{b_{jq}\}_{J \times Q}$ are the Mode B weights
- $\mathbf{C} = \{c_{kr}\}_{K \times R}$ are the Mode C weights
- $\mathbf{G} = \{g_{pqr}\}_{P \times Q \times R}$ is the 3-way **core array**
- $\mathbf{E} = \{e_{ijk}\}_{I \times J \times K}$ is the 3-way error array

Visualization of Tucker3 Model (Smilde et al., 2004)



PARAFAC/CANDECOMP Model Form

Let $\mathbf{X} = \{x_{ijk}\}_{I \times J \times K}$ denote a three-way data array.

- x_{ijk} is signal from i -th voxel at j -th time point of k -th subject

A **trilinear model** has the form

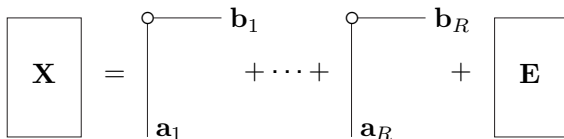
$$x_{ijk} = \sum_{r=1}^R a_{ir} b_{jr} c_{kr} + e_{ijk}$$

where

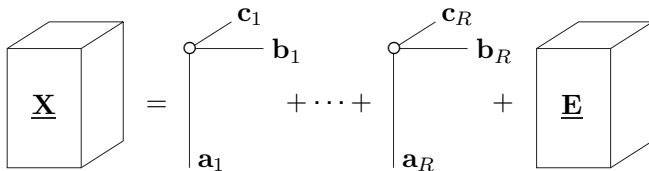
- $\mathbf{A} = \{a_{ir}\}_{I \times R}$ are the Mode A weights
- $\mathbf{B} = \{b_{jr}\}_{J \times R}$ are the Mode B weights
- $\mathbf{C} = \{c_{kr}\}_{K \times R}$ are the Mode C weights
- $\mathbf{E} = \{e_{ijk}\}_{I \times J \times K}$ is the 3-way error array

Visualization of PCA and Parafac (Helwig et al., 2012)

a) PCA: $\mathbf{X} = \sum_{r=1}^R \mathbf{a}_r \circ \mathbf{b}_r + \mathbf{E}$



b) Parafac: $\underline{\mathbf{X}} = \sum_{r=1}^R \mathbf{a}_r \circ \mathbf{b}_r \circ \mathbf{c}_r + \underline{\mathbf{E}}$



Relationship between Parafac and Tucker3

Parafac is a (very) special case of the Tucker3 model.

If $P = Q = R$ and if the core array $\mathbf{G}$ is **super-diagonal**:

$$g_{pqr} = \begin{cases} 1 & \text{if } p = q = r \\ 0 & \text{otherwise} \end{cases}$$

then Parafac and Tucker3 models are equivalent.

Can also consider Tucker3 model as a special case of Parafac model (see Harshman & Lundy, 1984, pp. 173–178).

Some More (Convenient) Notation for Parafac

Slice Notation:

- Let $\mathbf{X}_k = \{x_{ij(k)}\}_{I \times J}$ denote the k -th frontal slice of $\mathbf{X} = \{x_{ijk}\}_{I \times J \times K}$
- Let $\mathbf{E}_k = \{e_{ij(k)}\}_{I \times J}$ denote the k -th frontal slice of $\mathbf{E} = \{e_{ijk}\}_{I \times J \times K}$
- Let $\mathbf{C}_k = \text{diag}(c_{k1}, \dots, c_{kR})$ denote k -th row of $\mathbf{C} = \{c_{kr}\}_{K \times R}$
- Parafac Model: $\mathbf{X}_k = \mathbf{A}\mathbf{C}_k\mathbf{B}' + \mathbf{E}_k$

Stretch Notation:

- Let $\mathbf{X}_A = \{\mathbf{X}_1, \dots, \mathbf{X}_K\}_{I \times JK}$ denote the Mode-A matricized data
- Let $\mathbf{E}_A = \{\mathbf{E}_1, \dots, \mathbf{E}_K\}_{I \times JK}$ denote the Mode-A matricized error
- Let $\odot$ denote Khatri-Rao (i.e., columnwise Kronecker) product
- Parafac Model: $\mathbf{X}_A = \mathbf{A}(\mathbf{C} \odot \mathbf{B})' + \mathbf{E}_A$

Parafac Uniqueness (Intrinsic Axis Property)

Parafac is popular due to its **essential uniqueness property**.

- First suggested by Harshman (1970, 1972)
- Sufficient condition provided by Kruskal (1977)

Kruskal's sufficient condition for Parafac essential uniqueness:

- Define the ***k*-rank (Kruskal-rank)** of $\mathbf{Y}$ to be the largest integer k such that every subset of k columns of $\mathbf{Y}$ is linearly independent
- Suppose $(\mathbf{A}, \mathbf{B}, \mathbf{C})$ is a rank- R Parafac solution, and suppose that $k_{\mathbf{A}} + k_{\mathbf{B}} + k_{\mathbf{C}} \geq 2R + 2$, where $k_{(\cdot)}$ is k -rank of denoted matrix
- If $(\tilde{\mathbf{A}}, \tilde{\mathbf{B}}, \tilde{\mathbf{C}})$ is rank- R solution with the same error array $\mathbf{E}$, then

$$\tilde{\mathbf{A}} = \mathbf{A}\mathbf{P}\mathbf{D}_{\mathbf{A}}, \quad \tilde{\mathbf{B}} = \mathbf{B}\mathbf{P}\mathbf{D}_{\mathbf{B}}, \quad \tilde{\mathbf{C}} = \mathbf{C}\mathbf{P}\mathbf{D}_{\mathbf{C}}$$

where $\mathbf{P}$ is a permutation matrix and $(\mathbf{D}_{\mathbf{A}}, \mathbf{D}_{\mathbf{B}}, \mathbf{D}_{\mathbf{C}})$ are diagonal scaling matrices such that $\mathbf{D}_{\mathbf{A}}\mathbf{D}_{\mathbf{B}}\mathbf{D}_{\mathbf{C}} = \mathbf{I}_R$ (identity matrix)

Alternating Least Squares (ALS) Algorithm

The ALS algorithm is standard approach for estimating $(\mathbf{A}, \mathbf{B}, \mathbf{C})$ matrices (see Faber et al., 2003; Tomasi & Bro, 2006).

ALS Parafac Algorithm:

$$\textcircled{1} \quad \mathbf{A} = \mathbf{X}_\mathbf{A} \mathbf{Z} (\mathbf{Z}' \mathbf{Z})^{-1} \quad \text{where} \quad \mathbf{Z} = \mathbf{C} \odot \mathbf{B}$$

$$\textcircled{2} \quad \mathbf{B} = \mathbf{X}_\mathbf{B} \mathbf{Z} (\mathbf{Z}' \mathbf{Z})^{-1} \quad \text{where} \quad \mathbf{Z} = \mathbf{C} \odot \mathbf{A}$$

$$\textcircled{3} \quad \mathbf{C} = \mathbf{X}_\mathbf{C} \mathbf{Z} (\mathbf{Z}' \mathbf{Z})^{-1} \quad \text{where} \quad \mathbf{Z} = \mathbf{B} \odot \mathbf{A}$$

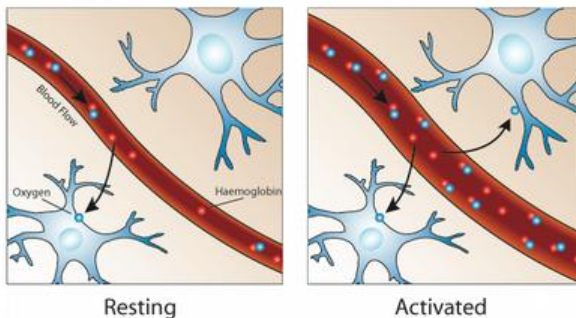
Some considerations for using ALS algorithm:

- Should try multiple random starts (of $\mathbf{B}$ and $\mathbf{C}$)
- Should try multiple values of R (unless true R is known)
- Poor performance when R is set too large or SNR is too small

Application to fMRI Data

Introduction to fMRI Data

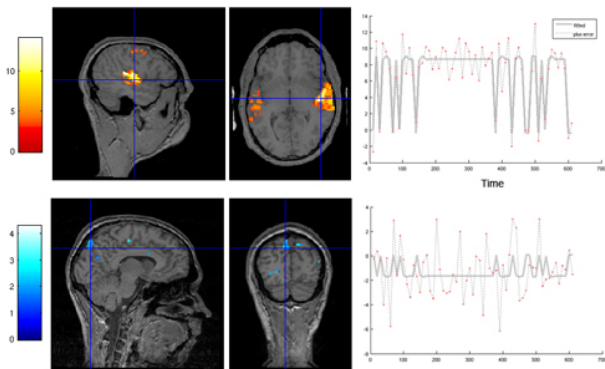
Functional magnetic resonance imaging (fMRI) measures brain activity by detecting changes in oxygenated blood flow.



From <http://www.bric.ed.ac.uk/research/techniques/functional.html>

BOLD Signal

Measures Blood-Oxygen-Level-Dependent (BOLD) signal to infer brain activation at a given location and time point.



From <http://www.bric.ed.ac.uk/research/techniques/functional.html>

Need for Unique fMRI Data Decomposition

Typical fMRI Data Decomposition Procedure

1. Register and aggregate signal
2. Select data to analyze (specific regions and/or time points)
3. Concatenate modes and apply PCA or ICA (or PICA)

Potential of Parafac Model

- Can analyze fMRI data from entire brain in one model
- Can obtain unique multimode decomposition
- Do NOT need to assume statistical independence

Tensor PICA Model

The Tensor PICA model (Beckmann & Smith, 2005) assumes that

$$\mathbf{x} = (\mathbf{C} \odot \mathbf{B})\mathbf{a} + \mathbf{e}$$

where

- $\mathbf{x} = (x_{11}, \dots, x_{JK})$ is observed JK -dimensional variable
- $\mathbf{a} \sim (\mathbf{0}, \mathbf{I}_R)$ is non-Gaussian latent **source signal** vector
- $\{\mathbf{C} \odot \mathbf{B}\}_{JK \times R}$ is the latent **structured mixing matrix**
- $\mathbf{e} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_{JK})$ is Gaussian latent **white noise** vector

Note: TPICA has same form as Parafac model, but now we assume that...

- Rows of $\mathbf{A}$ consist of R independent source signals
- Error array is composed of iid Gaussian noise

TPICA Parameter Estimation

To estimate $(\mathbf{A}, \mathbf{B}, \mathbf{C})$, Beckmann and Smith (2005) propose iterating between Hyvärinen's (1999) FastICA algorithm and Parafac ALS algorithm.

Two-Stage (iterative) TPICA Algorithm:

- 1 Estimate Source Signals
 - Use PICA to estimate $\hat{\mathbf{A}}$
 - Estimate compound (unstructured) mixing matrix $\mathbf{M}$
- 2 Mixing Matrix Decomposition
 - Assume $\mathbf{M} = \mathbf{C} \odot \mathbf{B}$ has Khatri-Rao structure
 - Let $\mathbf{M}_r$ denote r -th column of $\mathbf{M}$ written as $J \times K$ matrix
 - $\hat{\mathbf{b}}_r = \sqrt{s_1} \mathbf{u}_1$ and $\hat{\mathbf{c}}_r = \sqrt{s_1} \mathbf{v}_1$ where $s_1 \mathbf{u}_1 \mathbf{v}_1'$ is rank-1 SVD of $\mathbf{M}_r$

Some Critiques of TPICA

Theoretical Critiques

- Spatial independence assumption is very questionable for fMRI data
 - Why should we assume fMRI components are spatially independent?
 - fMRI data are sparse and nonnegative $\rightarrow$ independent = non-overlapping
- Spatial independence is NOT necessary to obtain unique decomposition
 - Parafac's intrinsic axis property holds regardless of spatial (in)dependence
 - Assuming independence may prevent us from obtaining true solution

Algorithmic Critiques

- TPICA algorithm is not guaranteed to converge
 - Iterates between two (unrelated) loss functions
 - Negentropy maximization and SSE minimization
- Spatial independence can and should be imposed simultaneously
 - Proposed by De Vos et al. (2012)

Beckmann and Smith's (2005) Simulation Results

Beckmann and Smith (2005) compared Parafac and TPICA when analyzing simulated fMRI data, and found that...

- TPICA outperformed Parafac when R was set too large
- TPICA outperformed Parafac when SNR was too small

Limitations of Beckmann and Smith's simulation study:

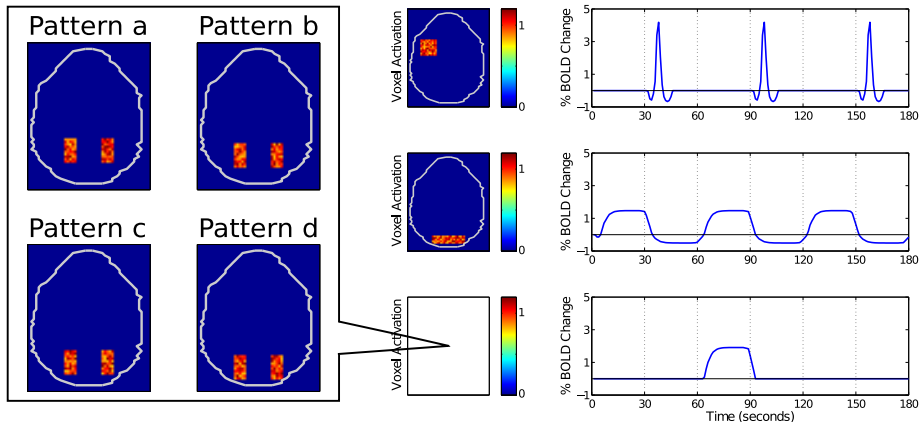
- Didn't compare TPICA and Parafac with correct (i.e., data-generating) R
- Didn't compare TPICA and Parafac with spatially dependent voxel maps
- Never considered Parafac with orthogonality constraint on voxel maps

To estimate Parafac with columnwise orthonormal Mode A weights use

$$\mathbf{A} = \mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1/2} \quad \text{where} \quad \mathbf{Z} = \mathbf{X}_A(\mathbf{C} \odot \mathbf{B})$$

in place of Step 1 of Parafac ALS algorithm (see Harshman & Lundy, 1994).

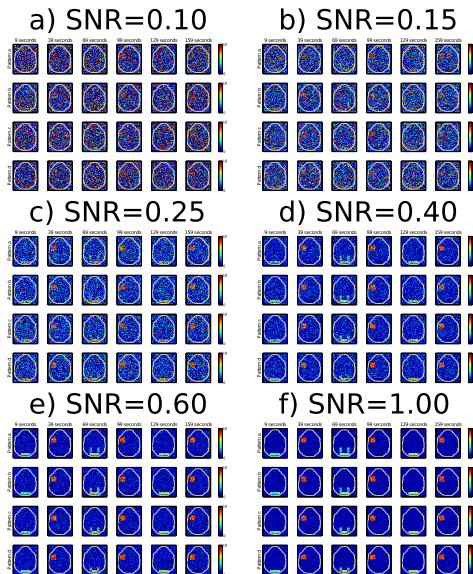
Some Simulated fMRI Data (Helwig & Hong, 2013)



Spatial pattern of activity differs across four patterns:

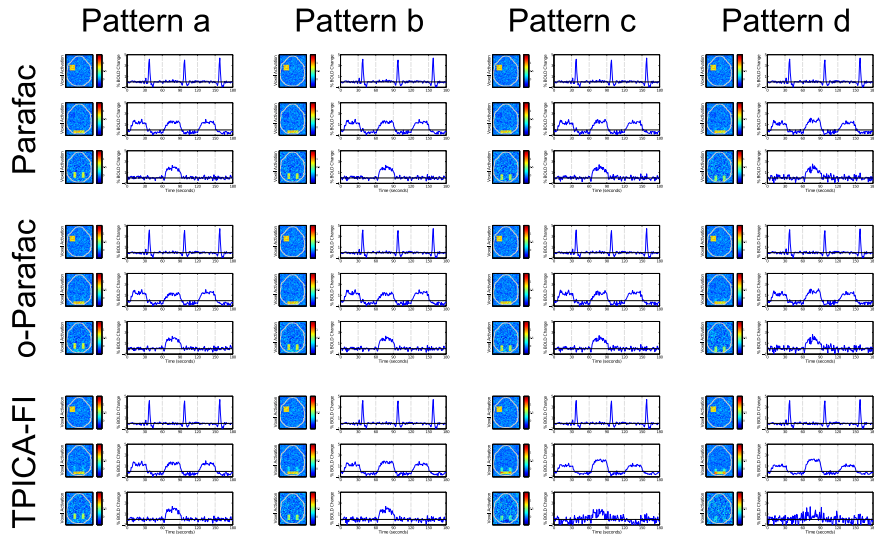
- (a) No overlapping activity
- (b) 10% overlapping activity
- (c) 30% overlapping activity
- (d) 50% overlapping activity

Simulated fMRI Data and SNR (Helwig & Hong, 2013)



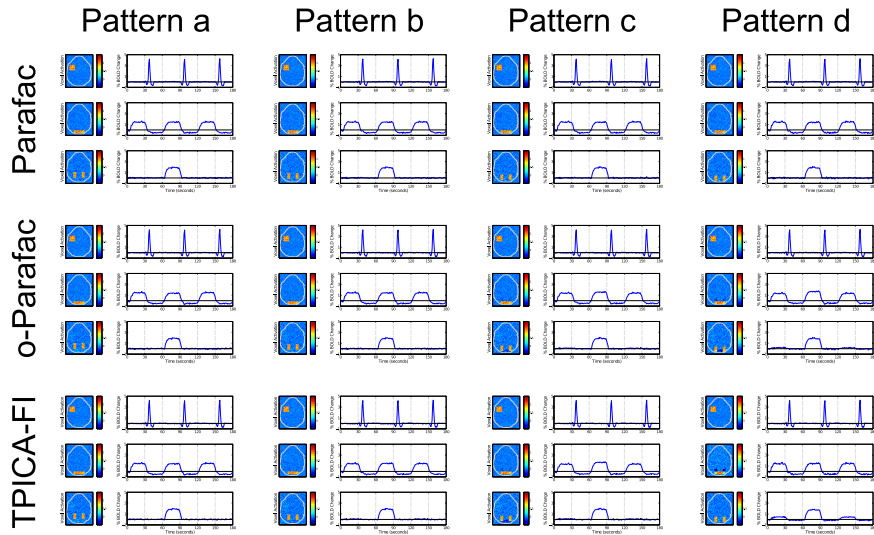
Parafac versus TPICA (SNR=0.10)

Figure 4 from Helwig and Hong (2013):



Parafac versus TPICA (SNR=0.25)

Figure 5 from Helwig and Hong (2013):



Parafac versus TPICA for fMRI Data Analysis

If spatial independence assumption is correct...

- TPICA outperforms Parafac at small SNR
- Orthogonal-Parafac performs as well as TPICA at small SNR

If spatial independence assumption is incorrect...

- TPICA produces noticeable bias for all SNR
- Orthogonal-Parafac produces smaller bias than TPICA

See Helwig and Hong (2013) for thorough critique of TPICA.

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